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RESEARCH ARTICLE

A Local Hierarchical LLM Framework for Privacy-Preserving Memory Forensics of Cryptocurrency Wallets

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ABSTRACT Cryptocurrency-related crime continues to expand worldwide. Chainalysis reports that the global value of illicit cryptocurrency transactions has exceeded USD 50 billion, underscoring an urgent need for more advanced digital forensics. Cryptocurrency investigations are particularly difficult when critical evidence resides in volatile memory. Traditional workflows are time-consuming and heavily manual. Cloud-based large language models (LLMs) also pose unacceptable privacy risks in confidential law-enforcement investigations. This paper presents a multi-layer reasoning framework that integrates LangChain with a locally deployed LLM (LLaMA 3.1-8B). The framework acquires volatile evidence via memory dumping, extracts forensic artifacts via keyword and regular-expression search, and performs three-stage reasoning with a Single-Layer Baseline Architecture, a Dual-Layer Supervisor Architecture, and a Tri-Layer RAG-Decider Architecture. We evaluate the framework on 100 purpose-built crypto-wallet forensic questions. The Tri-Layer architecture achieves an average human-evaluation total score of 11.29, which is an 8.9% improvement over the Single-Layer baseline. It also reaches a BERT F1 score of 0.84 in automated metrics, improving by 15.1%. Notably, the local Tri-Layer system performs very close to the commercial cloud model ChatGPT-4o (only a 0.3% gap overall) and surpasses it on the reasoning dimension. These results demonstrate that local LLM deployment can effectively support memory forensics under strict confidentiality and limited compute resources. The proposed approach offers a practical, low-cost, and privacy-preserving tool for digital investigations. It also shows that optimized lightweight local models can approach the analytical quality of cloud-scale models.

INDEX TERMS Cryptocurrency crime, digital forensics, memory forensics, LLM framework.

I. INTRODUCTION

Cryptocurrency-enabled crime continues to grow. Recent blockchain-analytics reports document sustained illicit activity, challenging law enforcement with fast-moving fund flows and limited actionable traces [1]. Criminal groups exploit the decentralization and anonymity of crypto wallets to conduct money laundering, fraud, and terrorist financing. In practice, disk forensics remains a standard procedure.

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However, scanning terabyte-scale storage is slow. This delay is problematic for cases that require timely disruption of illicit fund flows. In contrast, memory forensics captures volatile evidence while the system is running. This capability provides a key advantage for rapid on-site evidence acquisition.

Non-custodial wallets, represented by MetaMask, have reshaped the forensic landscape. Non-custodial wallets like MetaMask, with over 30 million active users [2], store secrets locally in browser extensions without server-side backups, making traditional recovery methods ineffective. The situation becomes even more challenging because suspects often

apply anti-forensic techniques. Common tactics include private browsing, clearing browsing history, locking the wallet, or restoring the system [3], [4]. These actions aim to erase digital traces. As a result, investigators must extract fleeting evidence from large, noisy, and unstructured memory images. This requirement has become a major bottleneck in modern digital forensics.

Prior studies have explored related directions. Van Der Horst et al. used the Volatility framework to scan memory of Bitcoin Core and Electrum clients [5]. Ali et al. extended memory forensics to analyze transactions in Bitcoin and Monero [6]. Many existing methods assume that investigators already possess key information such as private keys or wallet addresses. This assumption rarely holds in real crime scenes with limited intelligence. Current workflows also depend heavily on expert knowledge and manual filtering. This dependence reduces efficiency. It also causes inconsistent outcomes across investigators. End-to-end automated analysis and reasoning remain insufficient.

Large language models (LLMs) have strong capabilities in processing unstructured text and performing complex reasoning. These strengths create an opportunity for automated memory-data analysis. However, directly using cloud LLMs such as ChatGPT in law-enforcement forensics has a critical flaw. Forensic data contains sensitive case details and personal information. Uploading such data to third-party servers can violate confidentiality requirements and conflict with data-protection regulations. Therefore, there is an urgent need for local, automated forensic solutions that can operate in isolated environments.

To address these challenges, we propose a multi-layer reasoning system based on LangChain and a locally deployed LLM to support crypto-wallet memory forensics. To our knowledge, the core technical novelty of this work lies in three aspects that prior studies have not jointly addressed. First, we did not identify prior work that has embedded a locally deployed LLM into a complete memory-forensics pipeline—from volatile-memory acquisition through automated extraction and normalization to structured forensic reasoning—specifically for browser-based cryptocurrency wallets. Second, we design a three-tier generate-review-decide architecture (Single-Layer Baseline, Dual-Layer Supervisor, Tri-Layer RAG-Decider) that progressively suppresses hallucinations and strengthens evidence-chain reasoning, a mechanism that has not been reported in prior forensic LLM studies. Third, we demonstrate that this privacy-preserving local framework can achieve analytical quality comparable to a commercial cloud model (ChatGPT-4o) on multiple evaluation dimensions, suggesting that lightweight local deployment need not sacrifice forensic accuracy.

We further design a dual-track evaluation protocol combining six-dimensional expert review with ten automated metrics (including Jaccard, BERTScore, and ROUGE) to ensure objectivity and reproducibility.

We evaluate LLaMA 3.1-8B on 100 forensic questions spanning transaction patterns, anomaly detection, and smart-contract interactions. The Tri-Layer architecture consistently outperforms Single- and Dual-Layer baselines across human and automated metrics, approaches ChatGPT-4o overall, and exceeds it on reasoning coherence. These results confirm that well-designed multi-layer frameworks enable lightweight local models to achieve cloud-grade analytical quality.

The main contributions are summarized as follows:

- 1) **Memory-oriented multi-layer reasoning framework:** This work systematically applies LangChain-style supervision and collaboration mechanisms to crypto-wallet memory forensics. We design a three-tier architecture (Single-Layer Baseline, Dual-Layer Supervisor, Tri-Layer RAG-Decider). The framework reduces hallucinations and strengthens fact alignment and reasoning depth under noisy volatile-memory evidence.
- 2) **Lightweight local deployment with data sovereignty:** We deploy a lightweight LLM (LLaMA 3.1-8B) on a standard laptop with 32 GB RAM. The full workflow runs offline. This design preserves data sovereignty and case confidentiality. The proposed local system achieves performance close to the commercial cloud model ChatGPT-4o on multiple metrics. It also lowers adoption barriers for law enforcement in both security and cost.
- 3) **End-to-end forensic dataset and dual-track benchmark:** We design an eight-stage behavioral simulation that covers a wallet's lifecycle from installation to logout. This process generates representative memory-forensic data. We also build a 100-question professional test set and a dual-track evaluation protocol that combines six-dimensional expert review with ten automated metrics. This benchmark supports rigorous, reproducible comparisons in future research.

Overall, this work delivers a scalable, interpretable, and reproducible crypto-wallet memory-forensics pipeline. The pipeline integrates local LLM reasoning with structured evidence processing under strict confidentiality and limited resources. It provides direct practical value for law-enforcement investigations.

II. RELATED WORK

This section reviews related work in three areas: crypto-wallet digital forensics, LLM applications in forensic analysis, and retrieval-augmented generation frameworks.

A. DIGITAL FORENSICS OF CRYPTO WALLETS

Crypto-wallet forensics can be categorized by analysis layer: file system, memory, and wallet-specific artifacts. At the file-system layer, early studies focused on recovering static artifacts. Doran et al. [7] analyzed desktop wallets such as Multibit and Bitcoin-Qt on Windows and extracted key files (e.g., wallet.dat, db.log). Taylor et al. [8] and Kovalcik [9]

extended analysis to macOS and Linux, demonstrating cross-platform capability. However, these methods have limited ability to capture critical secrets (e.g., private keys) and transient states in non-custodial wallets. Many such secrets exist primarily in runtime memory.

Memory-layer analysis is more challenging but highly valuable. Van Der Horst et al. used Volatility to scan memory of Bitcoin Core and Electrum and identified addresses, private keys, and transaction IDs [5]. Ali et al. extended the approach to investigate malicious mining behavior [6]. Thomas et al. incorporated tools such as YARA and Cheat Engine to analyze memory persistence for hardware wallets [10]. These works confirm that volatile memory contains rich evidence for investigations.

Specialized research on browser wallets such as MetaMask is relatively limited but important. Mirza et al. conducted forensic examinations of Web3 wallets on Android and iOS [11]. Kim et al. focused on residual artifacts of browser extension-based crypto wallets [12]. Son et al. developed MetaMask-Forensics to parse LevelDB artifacts and to search seed phrases efficiently [13]. Compared with prior work, our study is the first to introduce a local LLM and a multi-layer reasoning framework into MetaMask memory analysis. We provide a local, multi-layer reasoning pipeline for browser-wallet memory forensics. Experiments show that local reasoning can analyze structured memory evidence with high accuracy. The approach reduces dependence on cloud compute and better balances privacy and efficiency for law-enforcement use.

B. LLMS IN DIGITAL FORENSICS

LLM applications in digital forensics are developing rapidly. For report generation, Michelet and Breitingner compared cloud and local LLMs for drafting forensic reports [14]. They found that cloud models such as ChatGPT often produce more accurate and complete text but still require human verification. Scanlon et al. examined the feasibility of LLMs for disk-image and memory-dump analysis and emphasized hallucination risks without human validation [15].

For memory analysis, Oh et al. proposed volGPT, which combines Volatility and ChatGPT to identify ransomware processes, reporting up to 94.1% accuracy [16]. Kim et al. proposed a Mobile Evidence Contextual Analysis (MECA) workflow that combines mobile forensic tools with an LLM to categorize multilingual chat logs and help police filter crime-relevant conversations [17].

For domain-specialized models, Sharma et al. proposed ForensicLLM based on LLaMA 3.1-8B, integrating RAG and RAFT to generate responses grounded in a forensic ontology [18]. However, many existing solutions rely on cloud APIs, which creates serious privacy risks for sensitive investigative data. In addition, prior work rarely targets crypto-wallet memory forensics with a dedicated multi-layer reasoning mechanism. To our knowledge, prior studies have not jointly integrated a locally deployed LLM into a complete

workflow spanning memory acquisition, automated extraction and normalization, and traceable forensic reasoning for browser-based cryptocurrency wallets. Table 1 provides a high-level comparative positioning of representative studies relative to the present work across four selected dimensions.

TABLE 1. Comparative positioning.

Study	Deployment	Layered Reasoning	Workflow	Wallet-Memory Focus
Van Der Horst et al. [5]	Local	-	Artifact analysis	Bitcoin Core
Oh et al. (volGPT) [16]	Cloud-assisted	-	Partial	-
Michelet & Breitingner [14]	Local / Cloud	-	Partial	-
Sharma et al. (ForensicLLM) [18]	Local	-	Partial	-
Kim et al. (MECA) [17]	Cloud-assisted	-	Partial	-
This work	Local	Yes	End-to-end	MetaMask

Note: This table is intended for high-level positioning rather than exhaustive evaluation of each study. Entries are based on the primary scope reported in the cited papers. A dash (-) indicates that the corresponding aspect was not a primary emphasis of the study. Michelet and Breitingner [14] evaluated both local and cloud-based configurations.

C. LLM DEVELOPMENT FRAMEWORKS AND RAG

LangChain is an open-source framework that standardizes and simplifies LLM application development [19]. Retrieval-Augmented Generation (RAG) is a key technique to mitigate LLM knowledge limits and hallucinations [20]. RAG retrieves relevant context from a trusted external knowledge base before generation. This design can substantially improve factual accuracy. RAG has been validated in high-stakes domains such as medical report generation [21] and mental-health assistance [22]. These successes suggest strong potential for forensic reporting and evidence analysis, where error tolerance is low.

However, existing LangChain and RAG use cases rarely address memory images. Memory dumps are highly fragmented and noisy. They also demand strong traceability. To fill this gap, we propose a multi-layer reasoning architecture that combines LangChain and a locally deployed LLM for systematic empirical evaluation in crypto-wallet memory forensics.

III. METHODOLOGY

The experimental pipeline involves two distinct stages executed in separate environments to ensure both evidence purity and computational efficiency.

Stage 1: Evidence Acquisition: All forensic simulations and memory dumps are conducted inside an isolated VMware

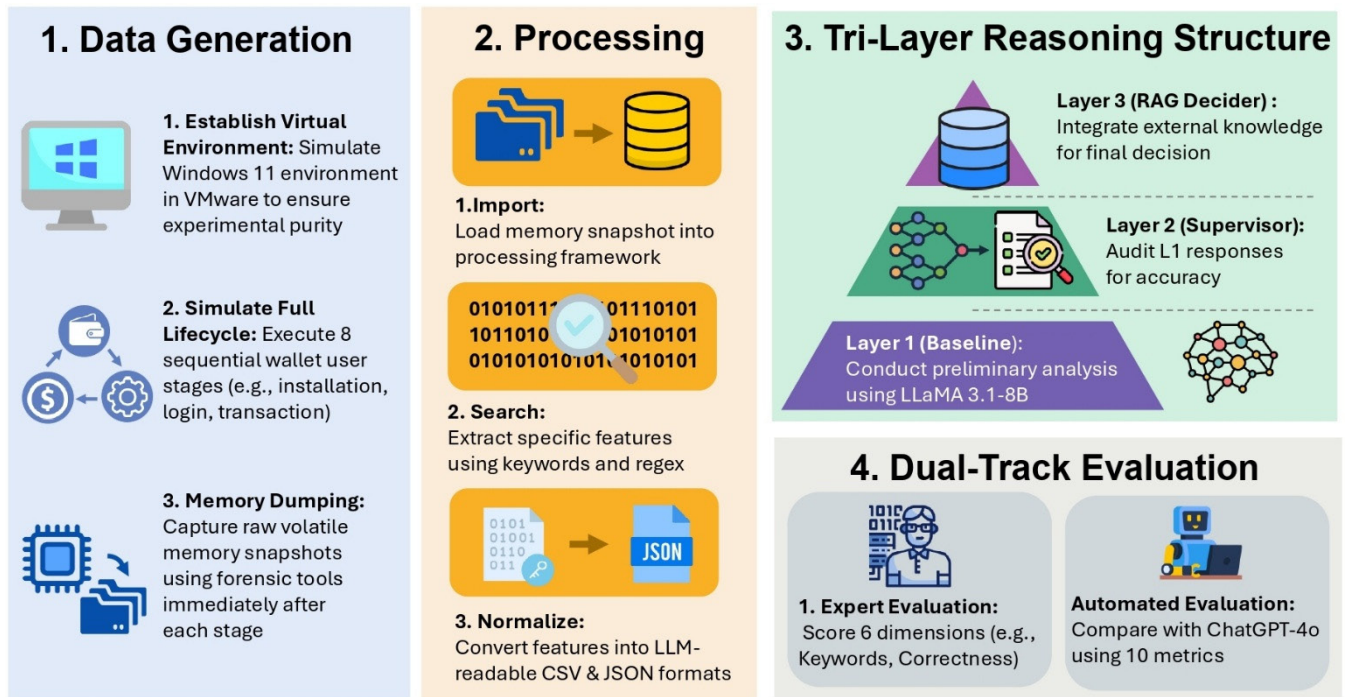


FIGURE 1. Evidence processing and multi-layer reasoning architecture.

virtual machine running Windows 11. The VM is allocated 12 GB RAM from the host system. Only essential software is installed, including Microsoft Edge (v131.0.2903.86), MetaMask (v11.16.14), and FTK Imager. This isolation ensures that the memory images reflect only the target wallet behavior without contamination from unrelated processes.

Stage 2: LLM Reasoning and Analysis: All LLM inference, LangChain orchestration, and RAG retrieval are executed on the host operating system with full access to the physical hardware: an 11th Gen Intel Core i7-11370H (4 cores / 8 threads) with 32 GB RAM. LLaMA 3.1-8B is served locally via Ollama with Q4_K_M quantization. No dedicated GPU is used; inference is entirely CPU-based. This two-stage design reflects a realistic deployment scenario where investigators acquire evidence on a target machine and perform analysis on their workstation.

Fig. 1 shows the end-to-end workflow. To ensure completeness and verifiability, we divide the method into four stages: Stage 1 (Data Generation), Stage 2 (Evidence Processing), Stage 3 (Tri-Layer Reasoning Structure), and Stage 4 (Dual-Track Evaluation). The scenario design follows prior work [23] and extends it with a multi-layer reasoning architecture and an evaluation protocol.

A. DATASET

To generate realistic memory traces, we simulate eight phases that cover a wallet's lifecycle. The phases include: installing MetaMask, importing a seed phrase and logging in, performing on-chain actions, transferring funds to a target wallet,

first lock/logout, rebooting the system, logging in again and displaying the private key, and finally logging out again. After each phase, we immediately acquire a full memory dump. This design captures volatile residues under different operations.

B. EVIDENCE PROCESSING

Although LLMs can process unstructured text, crypto-wallet memory images contain heavy binary noise, fragmented content, and lengths far beyond typical context windows. Directly feeding a memory image into an LLM increases cost and weakens traceability.

To focus the LLM on high-level semantic reasoning and evidence linking (instead of low-level string searching), we design an automated three-step pipeline. The pipeline transforms raw memory images into structured features that preserve forensic semantics. These features serve as input to the multi-layer reasoning system

- 1) Dump:** After each simulated phase, we use FTK Imager to acquire a full memory image of the virtual machine. The output is a binary *.mem* file.
- 2) Search:** We use WinHex text filtering to extract readable ASCII and Unicode strings from the binary image. We then run a custom Python script. The script uses a predefined keyword list (covering wallet software, major assets, and transaction signals) and regular-expression patterns (for Ethereum addresses, transaction hashes, and JSON-RPC fields). This step pinpoints wallet-related traces. Representative forensic dimensions and rules are summarized in Table 2.

TABLE 2. Memory forensics evidence search strategy for cryptocurrency wallets.

Forensic Dimension	Search Indicators (Keywords & RPC)	Precise Extraction Patterns (Regex)
Identity & Wallet Identify wallet presence, accounts, and network settings	Keywords: MetaMask, KeyringController, Secret Recovery Phrase, wallet, account RPC: eth_accounts, net_version	Address: "address"\s*:\s*"0x[a-fA-F0-9]{40}" Chain ID: "chainId"\s*:\s*"0x[a-fA-F0-9]{1}"
Transaction Activity Reconstruct fund transfers and transaction history	Keywords: transaction, value, confirmed, gas, transfer, swap RPC: eth_sendTransaction, eth_getTransactionReceipt	Tx Hash: "hash"\s*:\s*"0x[a-fA-F0-9]{64}" Value: "value"\s*:\s*"0x[a-fA-F0-9]{1}" Nonce: "nonce"\s*:\s*"0x[a-fA-F0-9]{1}"
Smart Contract Interaction Detect DApp/DeFi contract calls and approvals	Keywords: contract, token, approve, swap, ERC-20, function RPC: eth_call, eth_estimateGas	Input Data: "input"\s*:\s*"0x[a-fA-F0-9]{64}" Contract: "contractAddress"\s*:\s*"0x[a-fA-F0-9]{40}"
Asset & Token State Verify mainnet assets and token balances	Keywords: balance, symbol, nativeCurrency, ETH, USDT, USDC RPC: eth_getBalance, token query APIs	Balance: "balance"\s*:\s*"0x[a-fA-F0-9]{1}" Symbol: "symbol"\s*:\s*"0x[a-fA-F0-9]{2,10}"
Network & Security Events Trace node connections, errors, and risks	Keywords: rpc, infura, etherscan, error, reverted, phishing RPC: web3_clientVersion, eth_getLogs	Gas Price: "gasPrice"\s*:\s*"0x[a-fA-F0-9]{1}" Log Data: "data"\s*:\s*"0x[a-fA-F0-9]{64}" Error: "error"\s*:\s*"0x[a-fA-F0-9]{1}"

3) **Normalize:** We remove duplicates and malformed entries (e.g., incorrect length, encoding errors). We then extract key fields (e.g., timestamp, from, to, value, transactionHash) and output a unified CSV file. This structured CSV becomes the evidence input for LLM reasoning.

The full rule library includes over 120 keywords, 18 regex patterns, and 25 JSON-RPC method mappings. These mappings identify standard Ethereum-node interactions such as transaction broadcasting and balance queries. The rule set is extensible. It can be updated for different wallet versions or different blockchain-network environments.

C. MULTI-LAYER REASONING ARCHITECTURE

Base LLMs often suffer from hallucinations, inconsistency, and shallow reasoning in professional forensic tasks. We therefore design a progressive multi-layer reasoning architecture using LangChain. The core idea is to introduce supervision and collaboration to validate and correct reasoning steps. This design improves factual accuracy and logical rigor.

- 1) **Single-Layer Baseline Architecture:** This level uses a single LLaMA 3.1-8B model as a baseline. We prompt the model as a “crypto-wallet data analyst.” We strictly restrict reasoning to the provided CSV evidence. If no explicit evidence supports an answer, the model must reply “I don’t know.” This constraint minimizes speculation.
- 2) **Dual-Layer Supervisor Architecture:** This design adds a second LLM as a “supervisor.” The supervisor receives the full initial response from the first layer. It checks grammar and semantics. It also verifies whether reasoning goes beyond the evidence. It flags hallucinations and formatting inconsistencies. It then produces structured correction suggestions.
- 3) **Tri-Layer RAG-Decider Architecture:** This design adds a third LLM as a “decider” and integrates RAG. The

decider receives outputs from the first two layers. When deeper background knowledge or fact checking is needed, it retrieves relevant passages from a FAISS vector database constructed from a focused corpus of five documents (comprising technical texts and one instructional module), selected for their coverage of blockchain fundamentals, Bitcoin and Ethereum wallet operations, transaction semantics, and smart-contract standards. The final answer is generated by combining structured CSV evidence, the initial response, supervisor feedback, and retrieved RAG context.

D. EVALUATION PROTOCOL

We evaluate system performance using a dual-track evaluation design. All models are tested on the same 100-question forensic benchmark. Table 3 describes the benchmark structure and example questions.

The benchmark spans five dimensions across transaction, contract, and device layers. This coverage ensures comprehensive and representative evaluation. In addition, the benchmark deliberately includes several pairs of semantically related questions that share similar investigative scope but differ in phrasing or temporal granularity (e.g., Q8 and Q23 both address average transaction amounts; Q5 and Q20 both probe abnormal night-time activity with different time windows; Q79 and Q81 both examine cryptocurrency-related browsing records from different angles). These pairs function as consistency checkpoints: if a model produces contradictory or substantially divergent answers to closely related queries drawn from the same underlying evidence, this signals hallucination or reasoning instability. This design follows the cross-validation logic commonly used in forensic interviewing, where overlapping questions are posed to detect inconsistencies in a subject’s account.

Human evaluation scores each response on six dimensions (Table 4). Each dimension uses a 0–2 scale (0 = inadequate, 1 = partial, 2 = fully satisfactory). We evaluate

TABLE 3. Structure and examples of the evaluation question set.

Category	n	Example Questions
Basic Transaction Behavior: Analyze frequency, time distribution, and fund-flow patterns for anomaly detection	25	Is there abnormal late-night trading (00:00–06:00)?; Top-5 destination addresses by outgoing flow?; High-frequency trading (multiple tx within 10 minutes)?
Wallet and Contract Operations: Identify interactions with CEX/DeFi, risk-tagged addresses, and smart contracts	23	Interaction with risk-tagged addresses (gambling, laundering)?; Interaction with mixers (e.g., Tornado Cash)?; Has the user deployed any smart contract?
Gas and Transaction Anomalies: Detect abnormal fees, failures, and rapid "in-and-out" fund movements	20	Abnormal gas fees (higher than normal)?; Failed tx due to insufficient gas?; "Receive-and-transfer immediately" behavior?
Asset/Chain Usage: Assess concentration risk, stablecoin share, and cross-chain exposure	8	Which networks are mainly used (ETH/BSC/TRON)?; What is the stablecoin (USDT/USDC) share?; Which token is held or traded the most?
Device Records and Field Inference: Link extension artifacts with on-chain data to infer usage context	24	Was MetaMask installed and used?; Nonce gaps/anomalies?; Fund flows with blacklisted addresses?

Note: **n** denotes the number of questions assigned to each category; the total benchmark size is 100 questions.

100 questions across four systems: Single-Layer Baseline, Dual-Layer Supervisor, Tri-Layer RAG-Decider, and ChatGPT-4o (as a reference). This setup yields 400 responses.

All 400 responses are scored by a domain expert evaluator with operational experience in cybercrime investigation and digital forensics at a metropolitan police department. The evaluator scored each response independently against the structured CSV evidence and verified reference answers. Scoring followed a standardized protocol: for each question, the evaluator first reviewed the reference evidence in the CSV file, then read the model response, and assigned a score on each of the six dimensions according to the criteria in Table 4.

While the human evaluation relies on the consistent professional judgment of a single domain expert, we adopt a dual-track design to ensure broad methodological rigor. The human evaluation track captures professional judgment on forensic quality dimensions (e.g., reasoning coherence, hallucination detection) that automated metrics cannot fully assess. The automated evaluation track provides objective, reproducible verification through ten standardized metrics. Conclusions in this study are drawn only when both tracks converge. This complementary design ensures that no finding depends solely on one evaluation modality.

To reduce subjectivity and improve reproducibility, we also compute ten automated metrics for quantitative verification.

The automated evaluation track serves a different purpose from the human evaluation track. Its goal is not to

TABLE 4. Human evaluation dimensions and criteria.

Dimension & Definition	Scoring Criteria (0–2)
1) Keyword Matching Checks required forensic keywords/terms (e.g., hash, nonce)	2: all key elements covered 1: some missing 0: missed or off-topic
2) Correctness Verifies alignment with ground-truth memory evidence	2: fully correct 1: partially incorrect 0: severely incorrect
3) Reasoning Coherence Evaluates clarity and causality of reasoning steps	2: rigorous and clear 1: minor jumps but understandable 0: incoherent/no reasoning
4) Metric Alignment Checks correct use of requested quantitative metrics (e.g., Top-5, gas trend)	2: fully aligned 1: partially aligned 0: missing/misunderstood
5) Factual Grounding (Hallucination Check) Detects fabricated addresses/tx/contract interactions	2: none 1: suspicious/unverifiable claims 0: clear fabrication
6) Clarity & Fluency Assesses readability and correct technical terminology	2: clear and professional 1: understandable but awkward 0: hard to read

establish absolute correctness but to measure how closely local architectures approximate a well-established commercial benchmark. We select ChatGPT-4o as the reference point for automated metrics because it represents a widely recognized upper bound in general-purpose language understanding and has been used as a reference in prior LLM evaluation studies.

Importantly, ChatGPT-4o is not treated as ground truth for the overall study. The human evaluation track—where the expert evaluator scores each response against verified CSV evidence—serves as the primary credibility anchor. Automated metrics provide a complementary, reproducible quantitative layer. For example, when both tracks show that Tri-Layer outperforms Single-Layer and Dual-Layer, the convergence strengthens the conclusion beyond what either track could support alone. Conversely, the non-significant difference between Tri-Layer and ChatGPT-4o in human evaluation ($p = 0.790$) is consistent with the automated-metric finding, suggesting comparable performance under this benchmark.

We compute similarity between each local architecture output and the corresponding ChatGPT-4o output using ten metrics grouped into string similarity, semantic similarity, and content overlap.

We use ten metrics: string similarity (Jaccard, Sørensen), semantic similarity (BERT Similarity, BERTScore Precision/Recall/F1), and content overlap (ROUGE-1/2/L, TF-IDF).

To clarify the evaluation mechanism, we present three representative formulas: Jaccard (string overlap), BERTScore F1 (semantic similarity), and ROUGE-1 (content overlap).

The Jaccard coefficient is defined as:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} = \frac{|A \cap B|}{|A| + |B| - |A \cap B|} \quad (1)$$

Here, A is the token set of the model output, and B is the token set of the reference output (ChatGPT-4o). The range is $[0, 1]$. Higher values indicate better lexical overlap.

The BERTScore F1 is computed as:

$$F1 = 2 \cdot \frac{P \cdot R}{P + R} \quad (2)$$

$P(Precision)$ averages the maximum cosine similarity between each candidate token and all reference tokens. $R(Recall)$ averages the maximum cosine similarity between each reference token and all candidate tokens. In our results, the Tri-Layer architecture improves BERT F1 from 0.73 to 0.84, indicating stronger semantic matching.

ROUGE-1 Recall measures content overlap and is defined as:

$$\begin{aligned} \text{ROUGE} - 1_{\text{Recall}} \\ = \frac{\sum_{u \in \text{Unigrams}(R)} \min(\text{Count}_C(u), \text{Count}_R(u))}{\sum_{u \in \text{Unigrams}(R)} \text{Count}_R(u)} \end{aligned} \quad (3)$$

where u is a unigram token. The numerator counts overlapping unigrams (with clipping), and the denominator counts total unigrams in the reference.

E. MODEL CONFIGURATION AND PARAMETERS

We selected LLaMA 3.1-8B as the base model because this study prioritizes three constraints simultaneously: offline deployability on standard investigative hardware, acceptable forensic reasoning quality, and stable integration with an open-source local inference stack. Table 5 summarizes practical selection considerations for LLaMA 3.1-8B and other candidate models available at the time of our experiments; it reflects deployment feasibility criteria rather than an empirical cross-model benchmark.

TABLE 5. Model comparison.

Criterion	LLaMA 3.1-8B	Mistral-7B	Phi-3-mini-3.8B	Gemma-2-9B
Parameters	8B	7.3B	3.8B	9B
Context window	128K	32K	128K	8K
Ecosystem maturity	Extensive	Ready	Ready	Ready
Prior forensic application	ForensicLLM [18]	-	-	-

Note: All four candidate models are released under permissive or open-community licenses that permit local deployment and academic research use. Phi-3-mini is available in 4K and 128K context-length variants; the 128K variant is listed for capability comparison.

Compared with smaller local models (e.g., Phi-3-mini at 3.8B parameters), the 8B parameter scale offers more reliable instruction following and multi-step reasoning for complex forensic questions. Compared with larger local models (e.g., 70B-class), LLaMA 3.1-8B remains feasible for quantized CPU-based deployment on a 32 GB host system without a dedicated GPU, which is important for law-enforcement environments without specialized accelerators. LLaMA 3.1-8B also provides a 128K context window, which is advantageous for processing lengthy forensic evidence. The model has mature support in the Ollama/LangChain ecosystem, enabling one-command local deployment with optimized quantization. Critically, Sharma et al. [18] independently selected LLaMA 3.1-8B for their ForensicLLM system, confirming its suitability as a baseline for digital forensics research.

The model was therefore selected as a practical balance between reasoning capability, local hardware feasibility, and implementation maturity, rather than as an assertion that it is universally superior to all alternatives. While newer models continue to emerge, LLaMA 3.1-8B represents the most validated choice at the time of our experiments. The modular LangChain architecture allows straightforward replacement if a more efficient local model becomes available.

We deploy LLaMA 3.1-8B locally via Ollama with Q4_K_M quantization to eliminate data-exfiltration risk. Key generation parameters are: temperature = 1.0, top_k = 50, top_p = 1.0, and max_tokens = 4096. These settings were tuned in pilot tests to balance stability, factual accuracy, and diversity. We adopt role-based prompt engineering to enforce boundaries and output formats. Table 6 summarizes role definitions.

TABLE 6. Prompt engineering and role definitions in the multi-layer architecture.

Layer	Role and Objective	Design Rules and Core Strategies
Layer 1: Baseline Crypto-wallet data analyst	Provide initial answers to the 100-question list based on the forensic CSV data	Boundary control: restrict reasoning to provided evidence. Hallucination suppression: if evidence is insufficient, reply "I don't know."
Layer 2: Supervisor Forensic supervisor	Review Layer-1 answers and propose corrections using the original data and question list	Check grammar/semantics, evidence boundary violations, hallucinations, and consistency/formatting.
Layer 3: Decider Decision analyst with domain knowledge	Integrate outputs from Layers 1–2, the CSV evidence, and external domain documents via RAG	Emphasize cross-source analysis with RAG. Output must correspond to the 100-question list and provide structured conclusions and investigative suggestions.

This “generate–review–decide” design transforms single-step generation into a supervised and knowledge-augmented

reasoning pipeline. It directly targets hallucinations, reasoning gaps, and missing domain context.

IV. RESULTS AND DISCUSSION

We evaluate four systems under the dual-track protocol: Single-Layer Baseline, Dual-Layer Supervisor, Tri-Layer RAG-Decider, and ChatGPT-4o (reference).

A. HUMAN EVALUATION RESULTS

Human evaluators score each response on six dimensions using a 0–2 scale. The total score is the sum across dimensions (maximum 12). Table 7 summarizes results across the four architectures.

TABLE 7. Human evaluation results across architectures.

Dimension	Single-Layer	Dual-Layer	Tri-Layer	ChatGPT-4o
Keyword Matching	1.71	1.85	1.90	1.94
Correctness	1.74	1.80	1.88	1.89
Reasoning Coherence	1.75	1.77	1.91	1.90
Metric Alignment	1.74	1.78	1.86	1.84
Factual Grounding	1.74	1.82	1.89	1.88
Clarity & Fluency	1.69	1.82	1.85	1.87
Total	10.37	10.84	11.29	11.32

Note: Each dimension is scored on a 0–2 scale, averaged across the 100-question test set. The total is the sum of all six dimensions. Bold red indicates the best performance among local architectures.

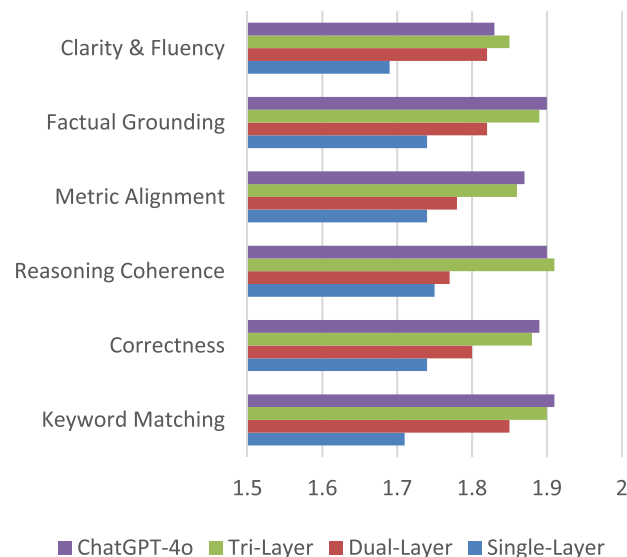


FIGURE 2. Visualization of human-evaluation results.

The Tri-Layer architecture reaches 11.29 (SD = 0.80, 95% CI [11.13, 11.45]), improving by 4.2% over Dual-Layer and 8.9% over Single-Layer. To validate the robustness

of these improvements, we perform paired statistical tests on per-question total scores. Table 8 reports paired t-test and Wilcoxon signed-rank test results with 95% confidence intervals.

TABLE 8. Statistical significance tests on total human scores.

Comparison	Mean Diff (Sig.)	t-test	Wilcoxon	95% CI
Tri vs. Single	+0.92 (***)	< .001	< .001	[0.63, 1.21]
Tri vs. Dual	+0.45 (***)	< .001	< .001	[0.20, 0.70]
Tri vs. GPT-4o	−0.03 (ns)	0.79	0.836	[−0.25, 0.19]
Dual vs. Single	+0.47 (**)	0.001	0.001	[0.20, 0.75]

Note: All p-values are two-tailed. Significance levels: *** $p < .001$; ** $p < .01$; * $p < .05$; ns = not significant.

The Tri-Layer architecture achieves statistically significant improvements over both Single-Layer and Dual-Layer ($p < 0.001$ in both parametric and non-parametric tests). The difference between Tri-Layer and ChatGPT-4o is not statistically significant ($p = 0.790$), indicating that no statistically significant difference was observed between the local system and the commercial cloud model under this benchmark.

Per-dimension Wilcoxon tests further show that Tri-Layer significantly outperforms Single-Layer in five of six dimensions: keyword matching ($p = 0.001$), correctness ($p = 0.013$), reasoning coherence ($p = 0.005$), Factual Grounding ($p = 0.009$), and clarity & fluency ($p = 0.016$). The only non-significant dimension is metric alignment ($p = 0.064$).

Notably, Tri-Layer scores 1.91 on reasoning coherence, slightly higher than ChatGPT-4o (1.90).

This is the only dimension where the local system exceeds the commercial model. The improvement is attributable to RAG retrieval of domain knowledge (e.g., Ethereum protocols and ERC standards). This retrieval compensates for the smaller model's limited knowledge base and strengthens evidence chaining and logical inference. The total-score gap between Tri-Layer (11.29) and ChatGPT-4o (11.32) is only 0.3%, indicating near commercial-grade performance.

B. AUTOMATED EVALUATION RESULTS

We compute ten metrics, grouped into string similarity, semantic similarity, and content overlap. ChatGPT-4o outputs serve as the reference. The automated evaluation results are summarized in Table 9.

Tri-Layer outperforms Single-Layer and Dual-Layer across all metrics. Its overall mean reaches 0.86, a 21.1% increase over Single-Layer. BERT F1 improves from 0.73 to 0.84 (+15.1%), indicating stronger semantic matching. TF-IDF increases from 0.71 to 0.86 (+21.1%), indicating better coverage of key evidence terms. Together with

TABLE 9. Automated evaluation results (Reference: ChatGPT-4o).

Metric Category	Metric	Single-Layer	Dual-Layer	Tri-Layer
String Similarity	Jaccard	0.72	0.80	0.86
	Sørensen	0.72	0.80	0.86
Semantic Similarity	BERT Similarity	0.71	0.81	0.85
	BERT Precision	0.72	0.79	0.86
	BERT Recall	0.69	0.81	0.88
	BERT F1	0.73	0.79	0.84
Content Overlap	ROUGE-1 F1	0.68	0.81	0.86
	ROUGE-2 F1	0.71	0.80	0.87
	ROUGE-L F1	0.72	0.81	0.86
	TF-IDF	0.71	0.80	0.86
Overall Mean		0.71	0.80	0.86

Note: Metrics are computed on the 100-question set using ChatGPT-4o outputs as reference (0–1; higher is better).

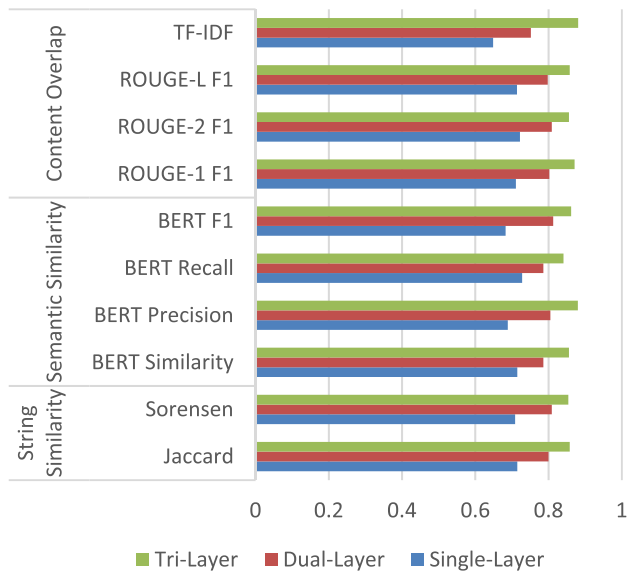


FIGURE 3. Visualization of automated evaluation results.

human evaluation, these results confirm the practical value and scalability of the proposed local multi-layer architecture.

C. DISCUSSION

Across both evaluation tracks, Tri-Layer shows consistent improvements over Single-Layer and Dual-Layer. The gains arise from three mechanisms:

- 1) Error correction by the supervisor layer: The second-layer supervisor intercepts common issues such as misread addresses, incorrect numeric values, and missing fields. This layer improves correctness and clarity and reduces hallucinations.
- 2) Knowledge augmentation by the RAG decider: The third-layer decider retrieves domain knowledge (e.g., ERC-20, ERC-721, ERC-1155). This retrieval yields the largest improvement in reasoning coherence and enables Tri-Layer to surpass ChatGPT-4o on this dimension.

- 3) Synergy of the three-stage collaboration: Single-Layer often stays at descriptive reporting. Dual-Layer improves form and correctness but remains limited without external knowledge. Tri-Layer forms a traceable reasoning chain (“generate–review–decide”). This chain produces outputs that are both evidence-grounded and professionally insightful.

To illustrate practical differences, we select representative questions from the 100-question set and compare response characteristics in Table 10.

Table 10 shows that Single-Layer tends to stay superficial, Dual-Layer adds key details, and Tri-Layer integrates external knowledge and multi-source evidence to produce investigation-oriented conclusions. This pattern supports the necessity of multi-layer reasoning and highlights the value of precise domain knowledge injection.

D. PRACTICAL IMPLICATIONS

The results show that even with a lightweight local model (LLaMA 3.1-8B) running on standard investigative hardware (i7 CPU, 32 GB RAM, no GPU), a carefully designed multi-layer reasoning mechanism can achieve analysis quality comparable to top commercial models. We discuss four concrete practical implications.

Data sovereignty and regulatory compliance: The full pipeline operates offline on a single workstation. No forensic data—including memory images, extracted artifacts, or reasoning outputs—leaves the investigator’s machine. This design satisfies strict data-handling requirements common in criminal investigations, where uploading case evidence to third-party cloud servers may violate agency policy, data-protection regulations, or chain-of-custody requirements.

On-site triage capability: As shown in Section IV-E, the end-to-end pipeline completes a single-case analysis in 3.0–3.5 hours under CPU-only inference. This turnaround makes the system suitable for on-site triage during search-and-seizure operations, where investigators need rapid initial analysis to guide further evidence collection. The lightweight hardware requirement (a standard laptop) enables deployment at field locations without specialized infrastructure.

Cost-effective adoption: All core components—Ollama, LangChain, FAISS, and the LLaMA model—are freely available. After initial setup, there are no per-query costs for LLM inference. For agencies handling substantial case volumes, this can reduce operational expenses compared to commercial cloud-based models.

Structured forensic output for judicial proceedings: The multi-layer architecture produces structured forensic narratives with professional terminology, quantitative indicators, and traceable evidence chains. These outputs can be integrated into formal case reports. The dual-track evaluation protocol provides a documented methodology for validating analytical quality, supporting the evidentiary admissibility of LLM-assisted forensic conclusions.

TABLE 10. Qualitative comparison on representative forensic scenarios.

Scenario Question	Single-Layer Traits	Dual-Layer Traits	Tri-Layer Traits
Q3 (Time distribution): What are the most frequent trading dates and time patterns?	Briefly notes activity concentrated on Apr 14 afternoon	Refines to 2–4 PM and notes no late-night tendency	Identifies strong afternoon regularity and discusses forensic meaning (e.g., scheduled tests or coordinated timing)
Q12 (Source of funds): Which address provides the most incoming funds?	Lists address 0xbdff...8228 as top source with 18 tx	Adds concentration insight (reliance on a single source)	Infers functional role (e.g., faucet or controlled internal account) with clearer forensic interpretation
Q26 (Core wallet identification): What is the primary wallet address used by the user?	Reports main address 0xe446...792e9	Cross-validates with internal wallet records	Builds a multi-evidence chain (tx count, total value, DApp interactions) to confirm dominance and role
Q35 (Risk-tagged interaction): Any interaction with risk-tagged addresses?	States lack of risk-tag fields prevents judgment	Restates limitation with template-like additions	Provides an explicit investigative procedure (e.g., cross-check with Chainalysis/Etherscan risk lists), demonstrating RAG value
Q63 (Illegal NFT interaction): Any interaction with illegal NFT projects?	Reports no NFT contract interaction observed	References ERC-721/1155 standards and confirms no interaction	Produces an exclusionary inference (initially ruling out illegal NFT involvement) with stronger logical closure
Q77 (MetaMask usage): Was MetaMask extension installed and used?	Confirms installation and usage	Adds state fields (isInitialized, isUnlocked)	Verifies functional completeness (asset transfer, signing) and builds a full functional evidence chain
Q87 (Suspicious contract detection): Do recipient addresses involve suspicious contracts?	Basic exclusion (normal addresses/contracts)	Notes most recipients are normal or common testnet contracts	Gives a stronger risk-exclusion statement with clearer evidentiary stance

E. DEPLOYMENT PERFORMANCE

To quantify the computational overhead of the multi-layer architecture, we profile inference latency and system resource consumption on the host hardware described in Section III (i7-11370H, 32 GB RAM, CPU-only, no GPU). Table 11 consolidates the deployment performance measurements.

TABLE 11. Deployment performance profile of the multi-layer architecture.

Metric	Single-Layer	Dual-Layer	Tri-Layer
LLM Calls per Question	1	2	3
RAG Retrieval	-	-	Yes
Mean Latency per Question (s)	29	51	102
Latency Ratio (vs. Single)	1.0×	1.8×	3.5×
Total Time for 100 Questions (min)	48	85	170

All values are measured on the host system described in Section III (i7-11370H, 32 GB RAM, CPU-only via Ollama). Actual performance may vary with hardware configuration, system load, and quantization settings.

The Tri-Layer architecture requires approximately 3.5× the latency of the Single-Layer baseline, consistent with three sequential LLM calls plus FAISS-based RAG retrieval per question. The Tri-Layer architecture remained feasible on a 32 GB CPU-only host system without a dedicated GPU. This overhead is operationally acceptable for offline forensic workflows, where analytical depth takes priority over speed.

The end-to-end pipeline from evidence acquisition to final analysis output comprises five stages: memory dump via FTK Imager (3–5 min, VMware guest), string extraction via WinHex (5–10 min, VMware guest), keyword and regex search in Python (1–3 min, host), CSV normalization (< 1 min, host), and Tri-Layer LLM reasoning (170 min, host CPU-only). Stages 1–4 show ranges because dump size and string density vary across memory images. A complete single-case analysis finishes within 3.0–3.5 hours under CPU-only inference, fully offline.

For agencies requiring higher throughput, GPU acceleration (e.g., NVIDIA RTX-class) could reduce per-question latency by several-fold. The modular LangChain architecture also permits base-model replacement without pipeline redesign, enabling future adoption of more efficient local models.

V. CONCLUSION AND FUTURE WORK

This paper proposed and evaluated a multi-layer LLM reasoning framework for privacy-preserving memory forensics of cryptocurrency wallets. The experimental results yield three key findings.

First, the progressive generate–review–decide architecture produces measurable quality gains. The Tri-Layer RAG-Decider achieves a human-evaluation total score of 11.29, a statistically significant improvement over Single-Layer (10.37, $p < .001$) and Dual-Layer (10.84, $p < .001$). These gains originate from supervisor-level error correction and RAG-augmented domain-knowledge injection, which jointly suppress hallucinations and strengthen evidence-chain reasoning.

Second, we did not observe a statistically significant quality difference between the local system and a commercial cloud model. The difference between Tri-Layer

and ChatGPT-4o is not significant ($p = .790$). On reasoning coherence, the local system (1.91) slightly exceeds ChatGPT-4o (1.90). These results suggest that a well-designed multi-layer architecture may compensate for the capacity gap between an 8B-parameter local model and a large-scale cloud model under this benchmark.

Third, the framework is operationally viable on standard hardware. The full pipeline completes a single-case analysis within 3.0–3.5 hours on a standard laptop (i7, 32 GB RAM, CPU-only). This confirms feasibility for offline, privacy-sensitive forensic workflows.

These findings should be interpreted in light of several limitations. In particular, evaluation is based on simulated scenarios, and extension to real-world multi-wallet, multi-chain cases requires further validation. Therefore, the findings should not be generalized to all wallet types or unconstrained forensic environments. Furthermore, while human scoring was conducted by a single domain expert and complemented by automated metrics, future studies should employ multiple independent raters to establish inter-rater reliability.

Our future work will prioritize testing the framework on anonymized memory images from closed cases under appropriate data-protection protocols, as well as conducting field trials that compare the system against existing investigator workflows. We also plan to extend the pipeline to support multi-wallet, multi-chain scenarios and to integrate cross-source reasoning with disk and network artifacts. The modular architecture is also applicable to other forensic domains where privacy-preserving local inference is required.

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Figure assets are from Flaticon.com.

DATA AVAILABILITY STATEMENT

The dataset used in this study was not provided by MetaMask or any third party. Instead, it was generated by the authors through controlled behavioral simulation in an isolated virtual-machine environment, as described in Section III-A. Raw memory images are not publicly released because they contain structurally realistic sensitive artifacts. The complete 100-question benchmark is not included in full due to space constraints, but can be shared upon reasonable academic request. Researchers wishing to reproduce the memory-acquisition stage may follow the simulation protocol outlined in that section using publicly available tools.

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